

AI assistant prototype

Module 6 - Lesson 2

Overview

Building an AI assistant prototype connects a model to a real product surface. Retrieval, prompts, and guardrails turn a raw model into something useful.

Key Concepts

- Define the assistant's scope and persona before writing prompts.
- Retrieval-augmented generation (RAG) grounds answers in your documents.
- Design prompts with clear instructions, context, and expected format.
- Chain the steps: retrieve relevant context, then generate a grounded answer.
- Add guardrails: refuse out-of-scope requests and cite sources.
- Evaluate responses with a small test set and iterate on prompts.

Example

```
def answer(query):  
    chunks = retrieve(query)          # RAG  
    prompt = build_prompt(query, chunks)  
    return call_model(prompt)         # grounded answer  
  
# Guardrail example  
if not is_on_topic(query):  
    return "I can only answer questions about this course."
```

Summary

An assistant prototype combines retrieval, prompt design, and guardrails. Ground answers in documents, keep scope tight, and evaluate responses to iterate toward reliability.

Practice Checklist

- Define the assistant's scope and a sample of test questions.
- Add retrieval so answers cite source content.
- Test five out-of-scope requests and confirm they are refused.